

Data Wrangling with Stata

Workshop 2 — Shaping data for analysis

Five operations: collapse, append, merge, reshape (wide and long).

Get the files:

`tinyurl.com/ndpop-stata-1`

green **Code** button → **Download ZIP** → unzip it → double-click `start_here.do`

Raw data is not a dataset



- **What you download is almost never what you analyze.**
- Three questions get you from one to the other:
 - What is the **unit of observation**? — what does one row represent?
 - What **variables** does the analysis need?
 - **Who is in the sample**? — which rows belong?
- **Today is mostly the first question.** The shape of the data.
- The rest — creating variables, correcting errors, and more — is a separate large effort.

Types of Data

- **Cross-section** (many units, once)
 - Static relationships between variables
- **Time series** (one unit, many periods)
 - Movement of a single unit over time
- **Repeated cross-section** (many units, over time — different units each round)
 - Individual dynamics are lost
 - If each round represents the same population, you can still study the dynamics of groups (e.g. states)
- **Panel / longitudinal** (the same many units, over time)
 - Dynamics at the unit level
 - Unobserved fixed traits of a unit can be canceled out

Where today's data sits

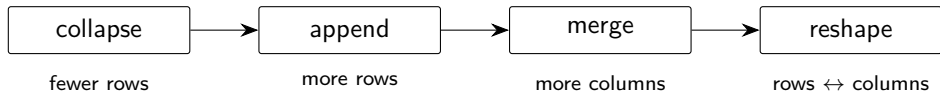
Three Census samples — 1980, 1990, 2000 — of *different people each time*.

A repeated cross-section, not a panel.

- You cannot follow a person from 1980 to 2000. Nobody appears twice.
- But *states* do appear in all three rounds.
- Aggregate to the state, and the group becomes the unit you can follow.

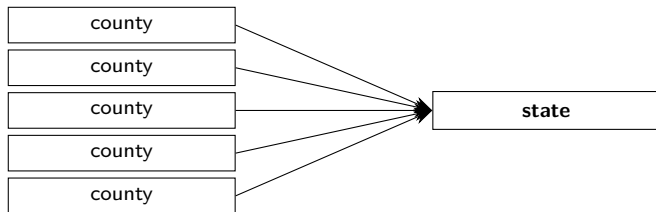
A note on geography: the Census suppresses counties it cannot identify into a single catch-all code. That is why today aggregates to the state.

Four operations, five blocks



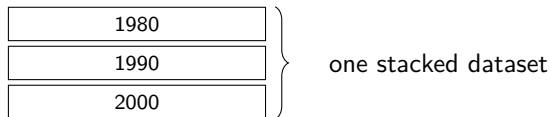
- Each block ends with a file. If yours does not work out, open the one provided and carry straight on.
- The handout is the instructions. This screen is only for the joins between blocks.

Block 1 — Collapse: aggregating the data



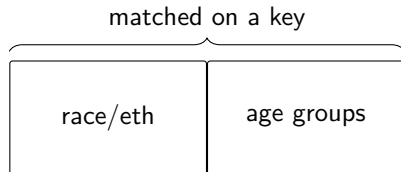
- Rows are **consumed**. 390 counties become 51 states.
- You must say *which statistic* — summed? averaged? the first row?
- Every variable you do not mention is **dropped**.

Block 2 — Append: stacking observations



- Same columns, more rows. The variables do not change.
- It works if the names and the *types* line up across files.
- A number in one file and text in another is the usual reason it fails.

Block 3 — Merge: joining on a key



Which side has repeats?

- **1:1** — one row meets one row
- **m:1** — many rows meet one
- **1:m** — one row meets many

Many-to-many needs a different tool.

Always check what matched.

- Matched both sides?
- Left over on the left?
- Left over on the right?

Unmatched rows are a finding, not a nuisance.

Keys have to match — in type, not just in value

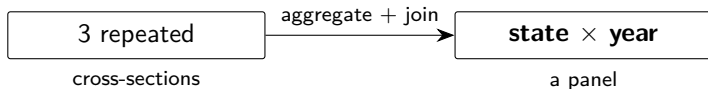
California is FIPS **06**.

Store that as a number and it becomes **6**.

The leading zero is gone, and it does not come back.

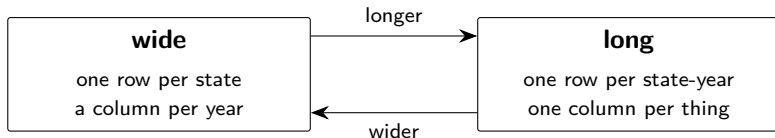
- Identifiers are labels, not quantities.
- A text key will not match a numeric one, however identical they look on screen.
- First thing to check when a merge looks like it should work.

You have just built a panel



- Same individuals are not available, but every state is.
- The *group* is the unit whose dynamics you can study.
- With the same units observed repeatedly — in the world of panel data.

Blocks 4 and 5 — Reshape: wide $\leftrightarrow$ long



- Same data. Same information. Different shape.
- Wide is sometimes easier for some tasks. Long is what most modelling wants.
- You must name the variable that identifies a row, and the one whose values become column names.
- Check that your row identifier really is unique.

What you did

The operations you ran:

- collapse — aggregated
- append — stacked
- merge — joined
- reshape — turned inside out

The habits worth keeping:

- Check the shape after every step
- Check what matched, every merge
- Confirm the key is a key
- Name files for what is in them

Everything today is in the take-home do-file, with every command filled in.